

Yucheng Zhang

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Education

The University of Texas at Austin

Aug. 2023 – May 2026

Bachelor of Science in Mathematics, Minor in Computer Science GPA: 3.67/4

Austin, TX

Relevant Courses: Data Structure(Python), Stochastic Process, Elements of Software Design, Data Analytics, Principles of Database System, Discrete Math, Real Analysis, Probability and Statistic, Elements of Networking, Accounting

Work Experience

Risk Analytics Intern (Continuing as Part-Time Analyst)

May. 2025 - Present

Toyota Industries Commercial Finance, Inc.

Dallas, TX

- Developed a full-stack credit risk modeling pipeline from raw SQL Server data, including ETL, feature engineering (e.g., PayNet inquiries, loan-to-value ratios, account tenure) and model training (Logistic Regression, Random Forest), improving default prediction accuracy by **3%** and reducing manual effort by **10%**.
- Validated models with statistical testing frameworks (ROC/AUC, feature importance, confusion matrices), achieving AUC >0.82 across Class 8 trucks, trailers, and used vehicle loans, and improving risk flagging accuracy by 3–4%.
- Conducted model **backtesting** by comparing predicted risk buckets against observed performance, revealing up to **3.7×** higher default rates in high-risk groups; findings informed refinements to Toyota's credit scorecard segmentation logic.
- Delivered results via Azure + Power BI, enabling **interactive risk dashboards**; applied advanced Excel reconciliation tools to align PayNet data with internal loan servicing systems for consistent reporting.
- Automated monthly model refreshes through parameterized SQL + Python workflows, dynamically updating high-risk customer lists for the DFS portfolio and reducing manual update time.

Financial Data Development Engineer – Fraud Detection

July. 2024 - Aug. 2024

Caitong Securities Ltd

Hangzhou, China

- Designed and optimized SQL-driven ETL pipelines processing 10M+ daily transactions, applying advanced SQL (window functions, subqueries, stored procedures) to improve extraction efficiency by **5%**.
- Implemented automated detection logic with parameterized thresholds and modular stored procedures, reducing false positives by **2%** and improving system reliability.
- Developed a customer risk scoring engine by integrating multiple data streams, enabling scalable evaluation of account behavior.

Software Engineering Intern – Customer Analytics

Nov. 2023 - June 2024

Joblogic-X corporation

Dallas, TX

- Engineered a data pipeline to scrape and process large volumes of social media (XHS) data, integrating with transactional order systems for scalable storage and analysis.
- Implemented ranking algorithms in Python with modular functions and clustering methods, generating prioritized customer lists with reproducible outputs.
- Integrated the KOC scoring module into Meet Fresh's AI-powered chatbot via **API**, improving system usability.
- Deployed and version-controlled codebase with Git, ensuring cross-platform accessibility and maintainability for future system extensions.

Data Scientist Assistant – Image Restoration

Jul. 2023 - Sep. 2023

Software Engineering Center of Chinese Academy of Science

Beijing, China

- Deployed and debugged image processing modules (VAE loss, Gaussian filters, LSGAN functions) on cloud servers via SSH, improving stability of large-scale archival photo/video restoration workflows.
- Integrated **GFPGAN API** into the pipeline for automated face restoration and color correction, reducing processing costs by **5%** and enhancing scalability for downstream applications.
- Version-controlled and containerized the project with Git for reproducibility and cross-platform deployment, while delivering a technical report benchmarking CNN against alternative models.

Research Experience / Projects

AI Architecture Contributor – CodeAssist Project

Aug. 2025 - Present

- Developed a modular architecture to integrate multiple AI providers, reducing dependency risk on a single vendor and supporting cost-efficient model selection.
- Enabled dynamic switching between **LLMs** (e.g., OpenAI, Hugging Face) to optimize performance vs. cost trade-offs in large-scale deployments.
- Contributed to **system design** that ensures long-term financial sustainability by lowering switching costs and preventing vendor lock-in.

Research Member – Computer Vision & Machine Learning (Quality Grading Project)

Aug. 2022 - May. 2025

- Contributed to research published in *Expert Systems with Applications* (Elsevier, 2025, IF = 7.5): “New kiwifruit quality grading methods based on image multi-feature fusion and ResTNet model”^[1]
- Built an end-to-end vision pipeline with multi-feature extraction (color, texture, shape, defects), enabling scalable fruit grading.
- Integrated ResTNet model with Grad-CAM visualizations, enhancing interpretability and adoption in agricultural quality control

AI-Powered Financial Analysis System

Jan. 2023 - May. 2024

- Extracted and analyzed **100** financial reports using the **FinGPT API** from EDGAR, storing data in a scalable data lake for efficient processing
- Implemented ETL processes to structure data from textual reports, identifying key financial metrics such as revenue, net income, and liabilities
- Created and fine-tuned **LLM** models using **LangChain** for financial data interpretation, integrating with Crew AI to enhance processing accuracy
- Utilized the **OpenAI API** to read and analyze financial reports, generating intelligent agents for automated data extraction.

Technical Skills

Languages: Python, SQL, R, SAS, Java, Linux, SPSS, HTML/CSS, MATLAB, Bash, Java, C, C++, NET, Perl

Frameworks: Numpy, Scipy, Matplotlib, Seaborn, Pandas, Scikit-learn, Pytorch, NLTK, Mongoose, LangChain

Tools: Tableau, Github, Git, Power BI, LINUX, Microsoft Office, Docker, Azure, GCP, VS Code, Hadoop, Spark, Splunk, Oracle, Excel, Visio, JIRA

